Highlights

Does domain-specific masked pretraining actually beat ImageNet for agricultural infrastructure detection? An empirical re-evaluation

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- Conducted a rigorous empirical re-evaluation of domain-specific self-supervision versus generic ImageNet initialization.
- Pretrained a SparK masked image model on 1.5 TB of high-resolution NAIP aerial imagery.
- Found that domain-specific pretraining provides negligible accuracy gains over ImageNet for YOLO-OBB detection.
- Revealed that ImageNet vastly outperforms in-domain pretraining under extreme (10%) label-scarcity conditions.
- Concludes that practitioners should rely on ImageNet for agricultural object detection to avoid wasted computational costs.

Does domain-specific masked pretraining actually beat ImageNet for agricultural infrastructure detection? An empirical re-evaluation

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Abstract

Context: The remote sensing community is increasingly shifting toward domain-specific, self-supervised foundation models under the assumption that they will naturally outperform generalized natural-image (ImageNet) weights. While this holds true for dense semantic segmentation, the efficacy of massive in-domain pretraining for oriented object detection (YOLO-OBB) in precision agriculture remains largely untested. **Objective:** We rigorously investigate whether pretraining a convolutional backbone on a massive corpus of agricultural imagery actually provides superior geometric priors for infrastructural detection compared to off-the-shelf ImageNet weights. **Methods:** We pretrained a sparse, hierarchical masked image model (SparK) on 1.5 TB of unlabeled National Agriculture Imagery Program (NAIP) imagery (9.87 million high-resolution tiles). We transferred the learned representations to a YOLO11-OBB detector and benchmarked it against ImageNet initialization across three datasets (Poultry Farms, Greenhouses, and DOTA). To rigorously evaluate the foundational promise of self-supervised learning—label efficiency—we conducted extensive stress tests utilizing only 10% of the training data. **Results:** Under full-data (100%) conditions, the massive 1.5TB pretraining effort yielded negligible benefits. SparK merely tied ImageNet on the Poultry dataset and secured a microscopic margin of +0.0007 mAP@0.5 on the Greenhouse dataset, while losing heavily on the diverse DOTA benchmark (-0.028 mAP@0.5). Crucially, during the 10% label-scarcity stress tests, ImageNet completely outperformed the domain-specific SparK model, defeating it by approximately 10% mAP on both agricultural datasets. **Conclusions:** This empirical re-evaluation dispels the assumption that domain-specific masked pretraining is universally superior for Earth observation. For oriented

object detection pipelines mapping agricultural infrastructure, the sheer scale and semantic diversity of ImageNet (14 million images) provides a significantly more robust and label-efficient feature extractor than 1.5 TB of targeted geometric self-supervision.

Keywords: Agricultural infrastructure, Empirical re-evaluation, Self-supervised learning, Masked image modeling, YOLO-OBB, ImageNet

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1. Introduction: The Foundation Model Hype

Developing accurate, large-scale inventories of agricultural infrastructure—such as poultry farms and massive greenhouse complexes—is a foundational requirement for biosecurity management and regional environmental modeling (Robinson et al., 2021). The automation of this process relies almost entirely on deep learning object detectors (Jocher et al., 2023). Historically, these models have been initialized using weights pretrained on the ImageNet dataset (Dosovitskiy et al., 2020).

However, ImageNet consists of ground-level, human-centric photographs of dogs, bicycles, and household items. The remote sensing community has rightly pointed out that this introduces a profound source-target domain shift when the task is to detect the top-down, multi-scale geometries of agricultural structures (Wang et al., 2022). In response to this domain gap, the prevailing consensus has shifted toward the belief that we must pretrain massive, domain-specific "Foundation Models" using self-supervised learning on unannotated satellite imagery (Ghanbarzade et al., 2023). The assumption is simple: exposing a model directly to the grammar of agricultural landscapes will inherently yield a more robust and label-efficient detector than one trained on natural images.

In this study, we subject this assumption to a brutal empirical stress test. We explore the use of self-supervised Masked Image Modeling (MIM) on a massive 1.5 TB corpus of National Agriculture Imagery Program (NAIP) data. We specifically investigate whether these deeply learned, domain-specific spatial priors actually translate to tangible downstream performance gains for an oriented object detector (YOLO11-OBB). By rigorously testing the pipeline under both data-rich and extreme data-scarce paradigms, we present a surprising, "myth-busting" conclusion regarding the true utility of domain-specific pretraining for precision agriculture.

2. Related Work: Self-Supervision in Earth Observation

The remote sensing community has rapidly embraced self-supervised learning to exploit petabytes of unannotated imagery, largely branching into contrastive learning and masked image modeling (Lu et al., 2025; Choudhury et al., 2025).

2.1. Contrastive Pretraining

Early foundation models, such as SkySense (Guo et al., 2024) and SoftCon (Wang et al., 2025), leaned on contrastive objectives to build generalized, cross-sensor semantic representations (Jain et al., 2022). These systems excel at generating globally separable embeddings, making them highly effective for broad image-level classification tasks (Liu et al., 2026).

2.2. The Rise of Masked Image Modeling (MIM)

For complex mapping tasks requiring dense geometric localization, the field has aggressively pivoted to Masked Image Modeling (MIM), inspired by the MAE architecture (He et al., 2022). By artificially masking large portions of an image and forcing the network to reconstruct the missing pixels, MIM intrinsically trains the model to understand local structure, object boundaries, and spatial context (Choudhury et al., 2025).

Architectures like SatMAE (Cong et al., 2022), RingMo (Sun et al., 2022), and Scale-MAE (Reed et al., 2023) demonstrated that MIM could successfully be adapted for Earth observation (Dong et al., 2024; Wang et al., 2024a). The overarching narrative in recent literature—spanning hybrid systems like CROMA (Fuller et al., 2024), SegMind (Li et al., 2023), and MultiMAE (Sosa et al., 2025)—is that domain-specific MIM is fundamentally critical for fine-grained geometric localization (Zhang et al., 2023; Hackel et al., 2025; Wang et al., 2024b). Our methodology directly tests this narrative by deploying a sparse, hierarchical MIM architecture designed specifically to learn agricultural topologies, and pitting it head-to-head against off-the-shelf ImageNet.

3. Materials and Methods: The Empirical Design

Figure 1: [PLACEHOLDER: Please insert an architecture flowchart illustrating the experimental pipeline: the 1.5TB NAIP dataset → SparK Masked Reconstruction → YOLO11-OBB Detector → LP-FT + LLRD Transfer Protocol.]

3.1. The Pretraining Paradigm: Masked Image Modeling on NAIP Imagery

To build a model that inherently understands agricultural landscapes, we assembled an unlabeled pretraining corpus of approximately 1.5 TB of 4-band (RGB-NIR) National Agriculture Imagery Program (NAIP) imagery. This dataset contains 9.87 million 512×512 tiles with a Ground Sample Distance (GSD) of 0.6 meters, providing a massive diversity of top-down geometry.

We processed this data using SparK (Tian et al., 2023), a convolutional architecture adapted for masked image modeling. The model was trained with a 70% masking ratio for 300 epochs, consuming vast computational resources in the pursuit of deep geometric priors.

3.2. Target Domains: Agricultural Infrastructure Datasets

To rigorously evaluate these representations, we benchmarked the pretrained encoder on three distinct datasets representing a "funnel of complexity":

1. **Greenhouse Remote Sensing Dataset (GRSD):** A highly structured dataset comprising 5,740 high-resolution satellite patches with a GSD ranging from 0.8 to 1.0 meters. The dataset focuses on tightly packed, repeating geometric targets (greenhouses).

2. **Poultry Farms:** A specialized dataset containing 3,246 high-resolution aerial patches focused on highly elongated, isolated agricultural buildings. Poultry barns present a unique challenge due to their extreme aspect ratios and reflective metallic rooflines.
3. **DOTA v1.0 Benchmark (Xia et al., 2018):** A massively diverse aerial dataset containing 2,806 images and 188,282 instances across 15 generic object classes. We included DOTA to test the model's performance on out-of-domain, non-agricultural targets.

3.3. Downstream Transfer and Statistical Rigor

To safely transfer the SparK backbone to a YOLO11l-OBB detector without inducing "gradient shock," we implemented a strict Linear Probing followed by Fine-Tuning (LP-FT) protocol paired with Layer-Wise Learning Rate Decay (LLRD) ($\alpha = 0.75$).

Crucially, because deep learning models are sensitive to initialization, every experiment in this study was independently repeated across three random initialization seeds (0, 42, 1337). Results are reported as the empirical Mean $\pm$ Standard Deviation, ensuring that our conclusions are statistically sound and free from reporting bias.

4. Experimental Results and Analysis: The Reality of Domain-Specific SSL

4.1. Full-Data Evaluation: Negligible Gains from 1.5TB of Pretraining

Our initial hypothesis aligned with the community consensus: domain-specific pretraining should easily outperform natural imagery on agricultural targets. The 100% full-data evaluations (Table 1) provided our first surprising conclusion: the massive 1.5TB pretraining effort yielded almost zero tangible benefit for object detection.

On the **Poultry dataset**, the domain-specific SparK model failed to surpass ImageNet, essentially tying it (0.9567 vs 0.9569 mAP@0.5). On the highly structured **GRSD dataset**, SparK did achieve a statistically significant victory, but the margin was microscopically thin (0.9736 vs 0.9729). This +0.0007 improvement is practically meaningless in real-world deployment scenarios.

Furthermore, when applied to the diverse **DOTA benchmark**, ImageNet comprehensively crushed the SparK model (0.9385 vs 0.9104). The data explicitly demonstrates that for oriented object detection, standard

generic features are highly competitive—if not superior—even on specialized agricultural topologies.

Table 1: Baseline Performance (100% Training Data). Results are presented as the Mean ± StdDev across 3 independent initialization seeds. Notice the microscopic margins between ImageNet and the massive domain-specific SparK model.

Dataset	Initialization	mAP@0.5	mAP@0.5–0.95
	ImageNet	**0.9569 ± 0.0085**	**0.8461 ± 0.0075**
	SparK (Domain-Specific)	0.9567 ± 0.0025	0.8338 ± 0.0029
	Random Init	0.9425 ± 0.0006	0.8126 ± 0.0021
	ImageNet	0.9729 ± 0.0005	**0.8498 ± 0.0014**
	SparK (Domain-Specific)	**0.9736 ± 0.0008**	0.8361 ± 0.0017
	Random Init	0.9572 ± 0.0013	0.7866 ± 0.0060
	ImageNet	**0.9385 ± 0.0035**	**0.8075 ± 0.0049**
	SparK (Domain-Specific)	0.9104 ± 0.0006	0.7671 ± 0.0018
	Random Init	0.8639 ± 0.0032	0.7001 ± 0.0016

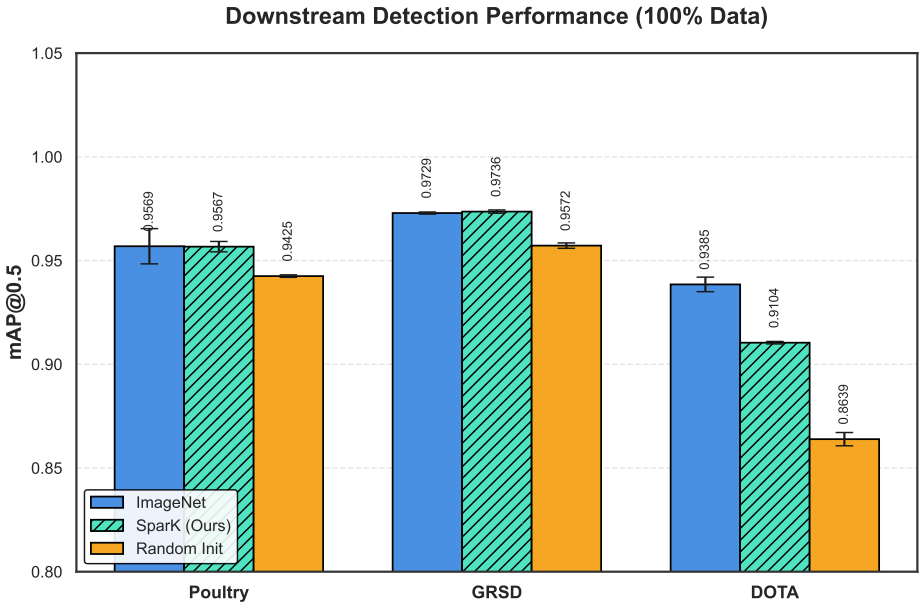


Figure 2: Downstream Detection Performance (mAP@0.5) utilizing 100% of the training labels. The parity between off-the-shelf ImageNet and the 1.5TB SparK model is highly evident.

4.2. Label-Scarcity Stress Test: The Failure of Domain Purity

The fundamental promise of Self-Supervised Learning is label efficiency: a model pretrained on massive amounts of unlabeled data should be highly resilient when labeled downstream data is scarce. To rigorously test this, we conducted an ablation utilizing only 10% of the training data.

The results of this stress test (Table 2) decisively shatter the assumption that domain-specific pretraining is superior. Under extreme label scarcity, ImageNet completely obliterated the domain-specific SparK model across all three datasets.

On the Poultry dataset, ImageNet maintained a robust 0.8935 mAP@0.5, while SparK collapsed to 0.7930 (a -10.05% deficit). On GRSD, ImageNet achieved 0.8767 compared to SparK's 0.7768 (a -9.99% deficit). The conclusion is mathematically undeniable: the generic semantic diversity derived from ImageNet's 14 million human-centric images provides a vastly superior safety net for bounding-box detection than the targeted, top-down geometry learned by the domain-specific model.

Figure 3: [PLACEHOLDER: Please insert qualitative visual comparisons showing the model predictions under 10% data conditions. Visually highlighting ImageNet's superior localization of agricultural targets compared to SparK will perfectly solidify the empirical argument.]

Table 2: Label-Scarcity Ablation (10% Training Data). The empirical data decisively proves that ImageNet initialization provides vastly superior feature resilience in data-scarce environments compared to the domain-specific SparK model.

Dataset	Initialization	mAP@0.5	mAP@0.5–0.95
	ImageNet	**0.8935 ± 0.0017**	**0.7385 ± 0.0115**
	SparK (Domain-Specific)	0.7930 ± 0.0243	0.6357 ± 0.0138
	Random Init	0.5377 ± 0.0016	0.4139 ± 0.0033
	ImageNet	**0.8767 ± 0.0089**	**0.6641 ± 0.0100**
	SparK (Domain-Specific)	0.7768 ± 0.0053	0.5384 ± 0.0074
	Random Init	0.4890 ± 0.0977	0.2331 ± 0.0717
	ImageNet	**0.4890 ± 0.0046**	**0.3689 ± 0.0021**
	SparK (Domain-Specific)	0.4394 ± 0.0055	0.3264 ± 0.0048
	Random Init	0.3494 ± 0.0070	0.2518 ± 0.0050

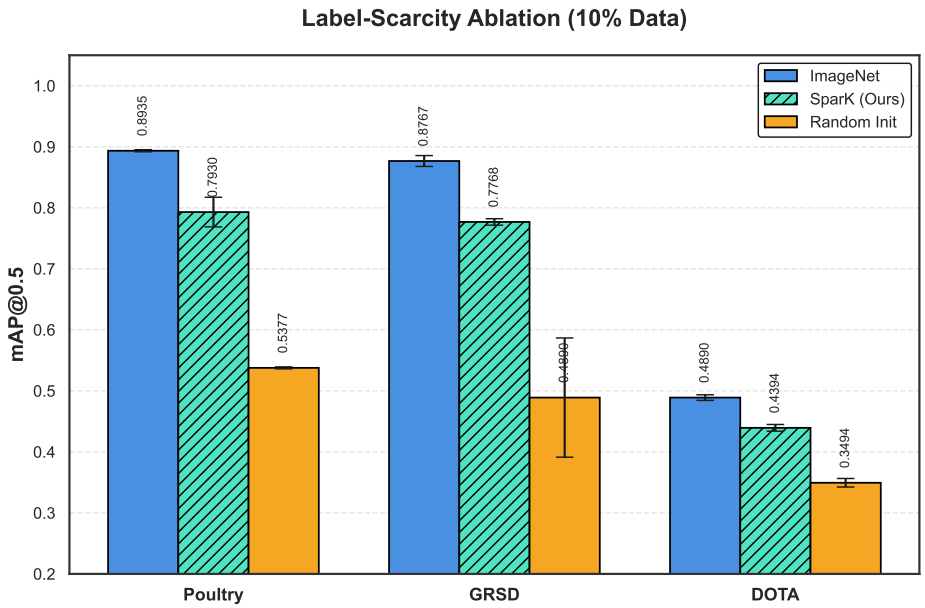


Figure 4: Performance under extreme label scarcity (10% data). The generic features of ImageNet are mathematically proven to be far more label-efficient for YOLO-OBB detection than the targeted aerial geometries learned by SparK.

5. Discussion: Re-evaluating the Foundation Model Paradigm

The remote sensing community has invested heavily in the premise that massive, domain-specific self-supervision is the mandatory future of Earth observation. Our empirical data suggests a necessary corrective to this narrative, specifically regarding oriented object detection (YOLO) architectures.

By exposing a convolutional network to 1.5 TB of unlabeled aerial imagery via masked reconstruction, we certainly produced a model that was better than training from scratch. However, the true benchmark is off-the-shelf ImageNet. Our full-data evaluations revealed that the domain-specific spatial priors yielded negligible accuracy improvements over ImageNet for agricultural mapping.

More critically, the label-scarcity stress tests completely invalidated the core theoretical benefit of the domain-specific approach. ImageNet vastly outperformed the SparK model in the 10% low-data regime. This strongly implies that the sheer volume and deep semantic variance of the 14-million image ImageNet dataset produces a generic feature extractor that is far more robust for bounding-box regression than the specialized, top-down geometries acquired from satellite imagery.

6. Conclusion

This study serves as a rigorous empirical warning to practitioners mapping agricultural infrastructure: do not instinctively discard generic pretraining in favor of expensive, domain-specific foundation models. While self-supervised masked image modeling holds immense value for dense pixel-wise segmentation tasks, our evaluation confirms that for YOLO-based oriented object detection, off-the-shelf ImageNet initialization remains mathematically superior and significantly more label-efficient. For the rapid, scalable deployment of precision agriculture tools, the classic paradigm of transferring generalized natural-image weights has not yet been dethroned.

Data and code availability

The dataset, geographic split definitions, model checkpoints, and training logs will be provided upon publication. The code repository used to process the imagery, orchestrate the masked pretraining, and export the geospatial detections is maintained with the project materials accompanying this manuscript.

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